ASSIGNMENT 10

Q1

B.C

$C_{A_1}(0) = C_L$, $C_{A_3}(L) = C_R$

$r_A = k_0$, $R_A = -k_0$

Diagram labels ϕ_1, ϕ_2

$C_2(l_1) = \phi C_1(l_1)$, $C_3(l_1 + l_2) = \phi_2 C_2(l_1 + l_2)$
$-D_1 C_1' = -D_2 C_2'(l_1)$, $-D_2 C_2'(l_1 + l_2) = -D_1 C_3'(l_1 + l_2)$

Governing equation

Medium 1 & 3 (no reaction)

$$\frac{d^2 C_1}{dx^2} = 0 , \quad \frac{d^2 C_3}{dx^2} = 0$$

$$0 = D_2 \frac{d^2 C_2}{dx^2} + R_A = D_2 \frac{d^2 C_2}{dx^2} - k_0 \Rightarrow \frac{d^2 C_2}{dx^2} = \frac{k_0}{D_2}$$

$C_1(x) = C_L + a_1 x$

$C_2(x) = \frac{k_0}{2D_0} \xi^2 + b_2 \xi + C_2$

$C_2(x) = C_R + a_3(x - R)$

At $x = l_1$ $(\xi = 0)$

Partition:

$$C_2 = C_2(l_1) = \phi_1 C_1(l_1) = \phi(C_L + a_1 l_1)$$

Flow continuity:

$$-D_1 a_1 = -D_2 b_2 \Rightarrow b_2 = \frac{D_1}{D_2} a_1$$

At $x = l_1 + l_2$ $(\xi = l_2)$

Flow continuity

$$-D_2\left(\frac{k_0}{D_2} l_2 + b_2\right) = -D_1 a_3$$

$$\Rightarrow D_1 a_3 = k_0 l_2 + D_2 b_2 = k_0 l_2 + D_1 a_1$$

$$C_3(l_1 + l_2) = C_R - a_3 l_3 = \phi C_2(l_1 + l_2)$$

$$C_2(l_1 + l_2) = \frac{k_0}{2D_2} l_2^2 + b_2 l_2 + C_2$$

substituting $b_2 = \frac{D_1}{D_2} a_1$

$$C_2 = \phi(C_L + a_1 l_1)$$

$$\therefore a_1 = \frac{C_R - k_0 l_2 l_3 - \phi\left(\frac{k_0 l_2^2}{2D_2} + \phi_1 C_L\right)}{l_3 + \phi_2\left(\frac{D_1}{D_2} l_2 + \phi l_1\right)}$$

$$C_1(x) = C_L + a_1 x$$

$$C_2(x) = \frac{k_0}{2D_2}(x-l_1)^2 + \frac{D_1}{D_2} a_1 (-l_1 + x) + \phi\left(C_L + a_1 l_1\right)$$

$$C_3(x) = C_R + \left(a_1 + \frac{k_0 l_2}{D_1}\right)(x - L)$$

$$J_{in} = -D_1 a_1$$

$$J_2(x) = -D_2 \frac{dC_2}{dx} = -\left[k_0 (x - l_1) + D_2 b_2\right]$$

$$J_{out} = J_{in} - k_0 l_2$$

Q2

B.C $\frac{dC_A}{dy} = 0$ $\frac{dC_B}{dy} = 0$

$D_A C_A'' + R_A = 0$

$D_B C_B'' + R_B = 0$

(a) $r_f = k_{f0}$ & $r_r = k_{r0}$

$R_A = -r_f + r_r = -(k_{f0} - k_{r0})$

$R_B = +r_f - r_r = k_{f0} - k_{r0} = -R_A$

$C_A'' = \frac{k_{net}}{D_A}$, $C_B'' = -\frac{k_{net}}{D_B}$

$k_{net} = k_{f0} - k_{r0}$

$R_A = -k_{net}$ $R_B = k_{net}$

$D_A C_A'' = R_A = -k_{net}$

$C_A = -\frac{k_{net}}{D_A}$

$C_A' = -\frac{k_{net}}{D_A} + C_1$

~~at~~ no flux at $y = L$

$\therefore C_1 = \frac{k_{net}}{D_A} L$

$$\therefore\ C_A' = \frac{k_{net}}{D_A}(L-y)$$

$$C_A = \frac{k_{net}}{D_A}\left(Ly - \frac{y^2}{2}\right) + C_2$$

Applying $C_A(0) = C_{A0}$

$$\therefore\ C_{A0} = C_2$$

$$C_A(y) = C_{A0} + \frac{k_{net}}{D_A}\left(Ly - \frac{y^2}{2}\right)$$

$$C_B'' = \frac{k_{net}}{D_B} \Rightarrow C_B' = \frac{k_{net}}{D_B}L + c_1 \Rightarrow c_1 = -\frac{k_{net}}{D_B}L$$

$$C_B' = \frac{k_{net}}{D_B}(y-L)$$

$$C_B = \frac{k_{net}}{D_B}\left(\frac{y^2}{2} - Ly\right) + c_2$$

using $C_B(0) = C_{B0} \Rightarrow c_2 = C_{B0}$

$$C_B(y) = C_{B0} + \frac{k_{net}}{D_B}\left(\frac{y^2}{2} + Ly\right)$$

(b)

$r_f = k_{f0}$ & $r_r = k_{r1} C_B$

$R_A = -r_f + r_r \quad = -k_{f0} + k_{r1} C_B$

$R_B = +r_f - r_r \quad = +k_{f0} - k_{r1} C_B$

$$D_B C_B'' = R_B$$

$$D_B C_B'' = k_{f0} - k_{r1} C_B$$

$$D_B C_B'' + k_{r1} C_B = k_{f0}$$

$$C_B'' + \frac{k_{r1}}{D_B} C_B = \frac{k_{f0}}{D_B}$$

$$\lambda^2 = \frac{k_{r1}}{D_B}, \quad \lambda = \sqrt{\frac{k_{r1}}{D_B}}$$

$$C_B'' + \lambda^2 C_B = \frac{k_{f0}}{D_B}$$, Now $C_B = C_{Bp}$, then $C_B'' = 0$

$$\therefore \lambda^2 C_{Bp} = \frac{k_{f0}}{D_B} \Rightarrow C_B = \frac{k_{f0}}{\lambda^2 D_B} = \frac{k_{f0}}{k_{r1}}$$

Homogeneous Solution

$$C_{Bh}(y) = A\cos(\lambda y) + B\sin(\lambda y)$$

General Solution

$$C_B = k_{f0} + A\cos(\lambda y) + B\sin(\lambda y)$$

Boundary Condition

$C_B(0) = C_{B0}$

$C_{B0} = \frac{k_{f0}}{k_{r1}} + A \cdot 1 + B \cdot 0 \Rightarrow A = C_{B0} - \frac{k_{f0}}{k_{r1}}$

No flux $y = L$ $C'_B(L) = 0$

$C'_B(y) = -A\lambda \sin(\lambda y) + B\lambda \cos(\lambda y)$

$0 = A\lambda \sin(\lambda L) + B\lambda \cos(\lambda L) \Rightarrow B = A \tan(\lambda L)$

$C_B(y) = \frac{k_{f0}}{k_{r1}} + A\left[\cos(\lambda y) + \tan(\lambda L) \sin(\lambda y)\right]$

or $C_B(y) = \frac{k_{f0}}{k_{r1}} + \left(C_{B0} - \frac{k_{f0}}{k_{r1}}\right) \frac{\cos(\lambda(L-y))}{\cos(\lambda L)}$

Now for $C_A(y)$

$D_A C_A'' = R_A = -k_{f0} + k_{r1} C_B(y)$

$C_B(y) = \frac{k_{f0}}{k_{r1}} + \left(C_{B0} - \frac{k_{f0}}{k_{r1}}\right) \frac{\cos(\lambda(L-y))}{\cos(\lambda L)}$

$$-k_{f0} + k_{e1} C_{B} = -k_{f0} + k_{e1}\left(\frac{k_{f0}}{k_{e1}}\right) + k_{e1}\left(C_{B} - \frac{k_{f0}}{k_{e1}}\right) * \frac{\cos(\lambda(L-y))}{\cos(\lambda L)}$$

$$R_A(y) = k_{e1}\left(C_{B0} - \frac{k_{f0}}{k_{e1}}\right)\frac{\cos(\lambda(L-y))}{\cos(\lambda L)}$$

$$\therefore C_A'' = \frac{R_A(y)}{D_A}, \quad \text{Let } M = \left(C_{B0} - \frac{k_{f0}}{k_{e1}}\right)\frac{1}{\cos(\lambda L)}$$

$$C_A' = -\frac{k_1}{D_A}\frac{M}{\lambda}\sin(\lambda(L-y)), \quad \boxed{C_1 = 0}$$

$$C_A = C_{A0} + \frac{D_B}{D_A} M\left[\cos(\lambda L) - \cos(\lambda(L-y))\right]$$

$$C_2 = C_{A0} + \frac{D_B}{D_A} M \cos(\lambda L)$$